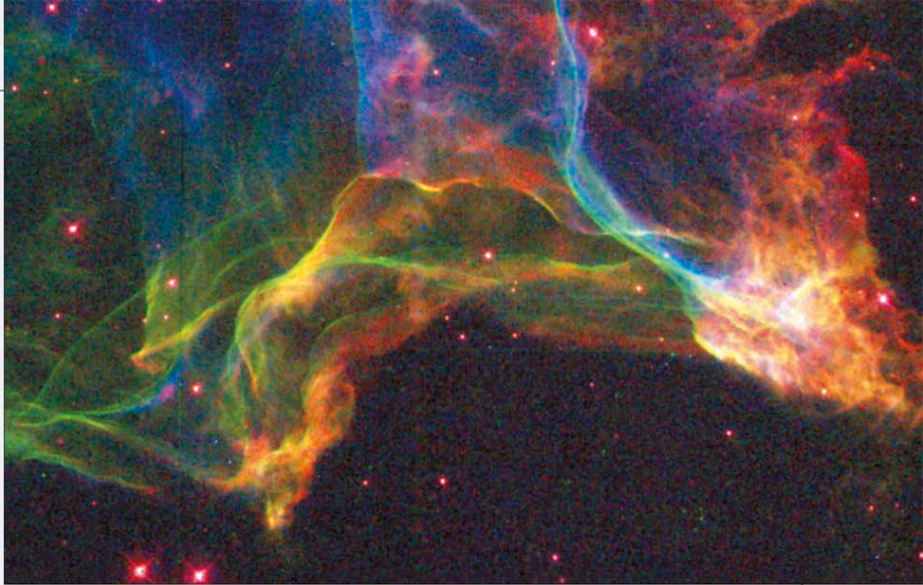


THE INTERSTELLAR MEDIUM

The interstellar medium, permeating the space between the stars, consists mainly of hydrogen in various states, together with dust grains. It constitutes about 10 percent of the mass of the Milky Way and is concentrated in the galactic disk. It is not distributed uniformly: there are clouds of denser material, where star formation takes place, and regions where material has been shed by stars, interspersed with areas of very low density. Within the interstellar medium, there is a wide range of temperatures. In the cooler regions, at around -440°F (-260°C), hydrogen exists as clouds of molecules. These cold molecular clouds contain molecules other than hydrogen, and star formation occurs where such clouds collapse. There are also clouds of neutral hydrogen (HI regions) with temperatures ranging from -280°F (-170°C) to $1,340^{\circ}\text{F}$ (730°C), and areas of ionized hydrogen heated by stars (HII regions) with temperatures around $18,000^{\circ}\text{F}$ ($10,000^{\circ}\text{C}$). Dust grains contribute about 1 percent of the galactic mass and are found throughout the medium. They are mostly small, solid grains 0.01 to 0.1 micrometers in diameter, consisting of carbon, silicates (compounds of silicon and oxygen), or iron, with mantles of water and ammonia ice or, in the cooler clouds, possibly solid carbon dioxide.



NONUNIFORM MEDIUM
As this image of the Cygnus Loop supernova remnant (see p.269) shows, material in the interstellar medium is very uneven. The blast wave from the supernova explosion is still expanding through the interstellar matter. Where it hits denser areas and slows down, atoms in the medium become excited and emit optical and ultraviolet light.

INVISIBLE COSMIC RAYS

Cosmic rays travel throughout the Milky Way. These are highly energetic particles that spiral along magnetic field lines. Cosmic rays are primarily ions and electrons and are an important part of the interstellar medium, producing a pressure comparable to that of the interstellar gas.

STARS

Stars are an important factor in the composition of the interstellar medium since they enrich the medium with heavy, metallic elements. A supernova explosion, the death of a massive star (see p.266), is the only mechanism that produces elements heavier than iron.

DARK NEBULAE

Dark nebulae are cool clouds composed of dust and the molecular form of hydrogen. They are only observed optically when silhouetted against a brighter background as they absorb light and reradiate the energy in infrared wavelengths. Stars are formed when dark nebulae collapse.

MAGNETIC FIELDS

Galactic magnetic fields are weak fields that appear to lie in the plane of the Milky Way, increasing in strength toward the center. They are aligned with the spiral arms but are distorted locally by events such as the collapse of molecular clouds and supernovae.

BETWEEN THE STARS

Contrary to early popular belief, the space between stars is not empty. The interstellar medium is fundamental in the process of star formation and galaxy evolution. Temperature defines the material's appearance and the processes occurring within it.

DUST CLOUDS

Young stars are often surrounded by massive disks of dust. These disks are believed to be the material from which solar systems are formed. There are often high levels of dust around stars in the later stages of their lives as they lose material to the interstellar medium.

REFLECTION NEBULAE

Material surrounding young stars contains dust grains that scatter starlight. In these nebulae, the density of the dust is sufficient to produce a noticeable optical effect. The nebulae appear blue because the shorter-wavelength, bluer light is scattered more efficiently.

EMISSION NEBULAE

When the interstellar medium is heated by stars, the hydrogen is ionized, producing a so-called HII region. The electrons freed by the ionization process are continually absorbed and reemitted, producing the red coloring observed in emission nebulae.